

Banking Transaction & Customer Behavior Analysis

SQL + Power BI Portfolio Case Study

Executive Summary

This project analyzes approximately 50,000 banking transactions recorded between 2020 and 2025. The objective was to transform raw transaction data into business-focused insights using MySQL for data validation, cleaning and analysis, followed by Power BI for interactive visualization.

The project was structured around 15 business questions covering transaction performance, customer segmentation, high-value activity, account behavior, transaction duration, login attempts, and account-level rankings. The final Power BI report contains two pages: an executive transaction overview and a customer/transaction behavior analysis page.

Business Problem

A banking organization needs a clear view of how transactions are distributed across time, transaction type, channels, customer segments and accounts. The project therefore focused on overall transaction performance, customer segmentation, high-value transaction activity, account-level performance, and trends over time.

Project Objectives

1. Validate and clean the transaction dataset.
2. Measure overall transaction volume and value.
3. Compare Debit and Credit behavior.
4. Identify yearly and monthly transaction patterns.
5. Segment customers by age and occupation.
6. Analyze transaction behavior by account balance and duration.
7. Identify high-value transactions using a \$1,000 analytical threshold.
8. Identify active and high-contribution accounts.
9. Examine high login-attempt activity.
10. Demonstrate advanced SQL and communicate the results through Power BI.

Dataset & Scope

The dataset contains approximately 50,000 transactions and approximately 500 unique accounts covering 2020–2025. Important fields included TransactionID, AccountID, TransactionAmount, TransactionType, TransactionDate, Location, Channel, CustomerAge, CustomerOccupation, AccountBalance, TransactionDuration and LoginAttempts.

Data Cleaning & Validation

The SQL workflow began with row-count and schema inspection. Transaction type, channel and occupation distributions were validated. A backup table was created before transformations.

Transaction dates were inspected using STR_TO_DATE and converted into DATETIME. TransactionAmount and AccountBalance were validated and converted to DECIMAL(12,2). Malformed source column names were cleaned. Account-level consistency checks were performed for CustomerOccupation and CustomerAge. Final validation compared the cleaned table with the backup and checked total rows and unique TransactionID values.

SQL Methodology

The analysis progressed from descriptive analysis to advanced analytical techniques. Foundational techniques included SELECT, COUNT, SUM, AVG, MIN, MAX, GROUP BY, ORDER BY and date functions.

CASE WHEN was used to create balance, age and transaction-duration segments. Advanced questions used Common Table Expressions (CTEs), subqueries and window functions. RANK() was used for account rankings, while PARTITION BY created separate rankings for Debit and Credit transactions.

Important Business Questions

- Q1. How many total transactions were recorded, and how many unique accounts were involved?
- Q2. What is the total transaction value, average transaction amount, minimum transaction amount, and maximum transaction amount?
- Q3. How do Debit and Credit transactions differ in transaction volume and total transaction value?
- Q4. Which occupation categories are associated with the highest transaction volume and transaction value?
- Q5. How have transaction volume and transaction value changed year-over-year from 2020 to 2025?
- Q6. Which months generate the highest and lowest transaction activity across the six-year period?
- Q7. How many high-value transactions exceed \$1,000, and what percentage of all transactions do they represent?
- Q8. Which accounts are the most active based on transaction frequency?
- Q9. How does transaction behavior differ across account-balance segments?
- Q10. How does transaction activity differ across customer age groups?
- Q11. Do longer transactions tend to have higher transaction values?
- Q12. Which accounts have transactions associated with unusually high login attempts?
- Q13. Which accounts generate more transaction value than the average account?
- Q14. How do all accounts rank based on their total transaction value?
- Q15. Which accounts generate the highest transaction value for Debit and Credit transactions?

Power BI Dashboard

Page 1 — Banking Transaction & Customer Behavior Analysis

- Total Transactions KPI
- Unique Accounts KPI
- Total Transaction Value KPI
- High-Value Transaction % KPI
- Annual Transaction Performance
- Debit vs Credit Performance
- Transaction Channel Performance
- Transaction Value by Age Group
- Date, Transaction Type, Channel and Age Group slicers

Page 2 — Customer Segmentation & Transaction Behavior

- Transaction Value by Customer Occupation
- Transaction Value by Transaction Duration
- High-Value Transaction Rate by Location
- High-Value Transaction Rate by Occupation

The dashboard was designed so the Power BI story directly reflects the SQL business questions.

Key Findings

1. Approximately 50,000 transactions and approximately 500 unique accounts were analyzed across 2020–2025.
2. Total transaction value was approximately \$14.9M.
3. Transactions above \$1,000 represented approximately 3.58% of all transactions.
4. The 41–60 age group generated the highest transaction value.
5. Long-duration transactions generated approximately \$6.9M, compared with approximately \$4.7M for Medium and \$3.4M for Short duration.
6. High-value transaction rates varied by location; Austin was approximately 11.43% in the dashboard analysis.
7. High-value transaction rates varied by occupation; Student was approximately 4.34%, Retired 3.48%, Doctor 3.26%, and Engineer 3.19%.
8. Account-level ranking showed substantial variation, with the highest-ranked account generating more than \$111K in transaction value.

Business Interpretation

Transaction value is concentrated across particular customer and transaction segments rather than being evenly distributed. Age, occupation and location analysis can help identify segments with higher transaction contribution or higher proportions of high-value activity.

The duration analysis highlights an operational pattern worth investigating further: longer transactions are associated with greater aggregate transaction value. Account-level rankings can support relationship management, while high-login-attempt analysis can provide a separate monitoring signal.

Business Recommendations

1. Prioritize high-contribution customer segments for targeted relationship and engagement strategies.
2. Monitor high-value transaction concentration by location and occupation.
3. Investigate why long-duration transactions are associated with higher transaction value.
4. Use account-level rankings to identify high-contribution accounts.
5. Review accounts with unusually high login attempts as a separate security-monitoring signal.
6. Consider transaction volume alongside high-value rate when prioritizing locations or occupations.

Technical Skills Demonstrated

SQL: data validation, data cleaning, date conversion, data-type conversion, aggregation, GROUP BY, CASE WHEN, CTEs, subqueries, RANK(), PARTITION BY and business-question design.

Power BI: KPI cards, slicers, trend analysis, categorical comparisons, customer segmentation, high-value rate analysis and dashboard design.

Analytics: business problem framing, exploratory analysis, segmentation, benchmarking, trend analysis and business insight generation.

Limitations

The dataset is observational, so relationships identified should not be interpreted as causal. The \$1,000 high-value threshold is an analytical threshold selected for this project. High-value rates should be considered alongside transaction volume. Login attempts are a monitoring variable, not proof of fraud or account compromise.

Conclusion

This project demonstrates an end-to-end analytics workflow: raw banking transaction data was validated and cleaned in MySQL, analyzed through 15 business questions, and translated into a two-page Power BI dashboard. The strongest portfolio aspect is the connection between the SQL questions and the Power BI story: transaction scale, customer segmentation, high-value behavior and account performance are presented as one coherent business analysis.